

Unified Source Theory (UST) v5

New Theorems T29–T32:

Compact Object Spectrum, Solar Fusion Tunnelling,
and Universal Time Unit

Niyazi ÖCAL

Independent Researcher, Ankara, Turkey

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Contents

1 Preamble: UST Core Constants

The universal constant is derived analytically:

$$N_{s,q} = \frac{3 - \sqrt{3}}{2} - \frac{\alpha}{17} = 0.63354460 \quad (1)$$

where $\alpha = 1/137.036$ and $17 = 4^2 + 1$ is the number of independent components of the 4D Dirac operator.

Dual-channel architecture:

$$N_b = 0.63354460 \quad (\text{Blueprint, Channel Q}) \quad (2)$$

$$Cc_b = 1 - N_b = 0.36645540 \quad (\text{Channel C}) \quad (3)$$

$$T_{Om} = e^{-2\pi N_b Cc_b} = 0.23252885 \quad (\text{tunnel amplitude}) \quad (4)$$

SU(3) identity (Theorem 4):

$$\frac{N_b}{Cc_b} = 1.728845 \approx \sqrt{3} = 1.732051 \quad (\Delta\% = 0.185) \quad (5)$$

2 Theorem 29 — Compact Object Spectrum

2.1 Statement

Compact objects form a spectrum parametrised by an effective UST constant N_{eff} :

White hole	$N_{\text{eff}} = 0.5$	symmetry centre
Star	$N_{\text{eff}} = Cc_b$	Channel C
Neutron star	$N_{\text{eff}} = T_{Om}$	equilibrium (T24)
Black hole	$N_{\text{eff}} = N_b$	Channel Q

(6)

2.2 Omnium Horizon Ratio

The Omnium horizon $r_{Om} = r_S/N_{\text{eff}}$ gives:

$$\frac{r_{Om}^{\text{white}}}{r_{Om}^{\text{black}}} = \frac{1/Cc_b}{1/N_b} = \frac{N_b}{Cc_b} = 1.728845 \approx \sqrt{3} \quad \Delta\% = 0.185 \quad (7)$$

2.3 Symmetry Breaking

At $N_b = Cc_b = 0.5$ (Big Bang pre-state):

$$T_{Om}(0.5, 0.5) = e^{-\pi/2} = 0.207880 \quad (8)$$

The symmetry breaks to $N_b = 0.6335$, leaving an imprint:

$$\Delta T_{Om} = T_{Om}(N_b) - T_{Om}(0.5) = 0.024649 \quad (9)$$

3 Theorem 30 — Solar Fusion Tunnelling

3.1 Statement

The Sommerfeld parameter η governing proton-proton quantum tunnelling in the Sun is:

$$\boxed{\eta = 6\pi \cdot N_b \cdot C_{c_b}} \quad (10)$$

3.2 Verification

$$\eta_{\text{UST}} = 6\pi \times 0.63354460 \times 0.36645540 = 4.376223 \quad (11)$$

$$\eta_{\text{obs}} = 4.383722 \quad (12)$$

$$\Delta\% = 0.171\% \quad \checkmark\checkmark\checkmark \quad (13)$$

The Gamow exponent follows directly:

$$\text{Gamow} = -2\pi\eta = -12\pi^2 \cdot N_b \cdot C_{c_b} = -27.4966 \quad (14)$$

$$\text{Observed} = -27.5437 \quad (15)$$

$$\Delta\% = 0.171\% \quad \checkmark\checkmark\checkmark \quad (16)$$

3.3 Physical Interpretation

The factor $6\pi = 2\kappa_2$ ($\kappa_2 = 2\pi N_b$ is Gear 2) connects solar fusion to the dual-gear structure. Solar energy production is Channel C $\rightarrow$ Channel Q tunnelling at rate $N_b \times C_{c_b} \approx T_{Om}$.

4 Theorem 32 — Universal Time Unit

4.1 Statement

The natural UST time unit is:

$$\boxed{1 \text{ UST-second} = N_b \times 1 \text{ SI-second} = 0.63354460 \text{ s}} \quad (17)$$

In this unit, the stabilisation rate becomes exactly:

$$\kappa \cdot dt_{\text{UST}} = \pi N_b \cdot \frac{1}{N_b} = \pi \quad (\text{exact, parameter-free}) \quad (18)$$

4.2 Cosmological Connection

$$t_H \times N_b = \frac{N_b}{H_0} = 9.191984 \text{ Gyr} \quad (19)$$

$$\nu_{\text{Cs-133}} = 9.192632 \text{ GHz} \quad (20)$$

$$\Delta\% = 0.0070\% \quad \checkmark\checkmark\checkmark \quad (21)$$

where $H_0 = 67.4 \text{ km/s/Mpc}$ (Planck 2018).

4.3 Physical Interpretation

The Cs-133 hyperfine frequency (atomic physics, micro scale) equals the Hubble time times N_b (cosmology, macro scale). This is a direct micro-macro bridge via the UST constant, extending the result of Theorem 7.

5 Summary Table

Table 1: New theorems T29–T32: predictions vs observations

Theorem	Content	Predicted	Observed	$\Delta\%$
T29	White/black hole ratio $\approx \sqrt{3}$	1.728845	$\sqrt{3} = 1.732051$	0.185
T30	Solar $\eta = 6\pi N_b C c_b$	4.376223	4.383722	0.171
T32a	$\kappa \cdot dt_{\text{UST}} = \pi$	π (exact)	π (exact)	0.000
T32b	$t_H \times N_b \approx \nu_{\text{Cs-133}}$	9.191984 Gyr	9.192632 GHz	0.007

Zero free parameters. All predictions derived from $N_{s,q} = 0.63354460$ alone.

6 Falsification

T32b predicts:

$$\frac{H_0 \cdot \nu_{\text{Cs-133}}}{N_b} = \text{constant} \approx 1.0 \quad (22)$$

If a more precise measurement of H_0 shifts this ratio beyond 0.1%, Theorem 32 must be revised.

References

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